

模电第五章作业 5.9-5.14 部分

(注：第五章前面的部分已用纸质版提交)

$$5.9. (a). \dot{F}_{iu} = \frac{\dot{I}_t}{\dot{U}_o} = -\frac{1}{R_t}$$

$$\dot{A}_{ust} = \frac{\dot{U}_o}{\dot{U}_s} = \frac{\dot{U}_o}{\dot{I}_s R_t} = -\frac{1}{R_t R_s}$$

$$(b) \dot{F}_{ui} = \frac{\dot{U}_t}{\dot{U}_o} = \frac{R_1}{R_1 + R_4}$$

$$\dot{A}_{ust} = \dot{A}_{uf} = \frac{\dot{U}_o}{\dot{U}_i} \approx \frac{\dot{U}_o}{\dot{U}_t} = \frac{R_1 + R_4}{R_1}$$

$$(c) \dot{F}_{ii} = \frac{\dot{I}_t}{\dot{I}_o} = \frac{-R_2}{R_2 + R_1}$$

$$\dot{A}_{ust} = \frac{\dot{U}_o}{\dot{U}_s} \approx \frac{\dot{I}_o R_L}{\dot{I}_t R_s} = \frac{1}{\dot{F}_{ii}} \cdot \frac{(R_4 // R_L)}{R_s} = -\frac{(R_1 + R_2)(R_4 // R_L)}{R_s R_2}$$

$$(d) \dot{F}_{ui} = \frac{\dot{U}_t}{\dot{I}_o} = \frac{-R_2 R_9}{R_2 + R_4 + R_9}$$

$$\dot{A}_{ust} = \frac{\dot{U}_o}{\dot{U}_i} \approx \frac{\dot{I}_o (R_7 // R_8 // R_L)}{\dot{U}_t} = \frac{1}{\dot{F}} (R_7 // R_8 // R_L)$$

$$= -\frac{R_2 + R_4 + R_9}{R_2 R_9} (R_7 // R_8 // R_L).$$

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5.12 由题意得

$$A_{sm} = 20 \lg 10^4 = 80 \text{ dB}$$

高频段三个截止频率为

$$f_{H1} = 10^4 \text{ Hz}, f_{H2} = f_{H3} = 10^5 \text{ Hz}$$

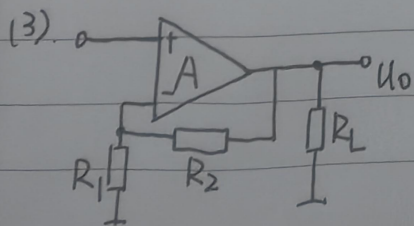
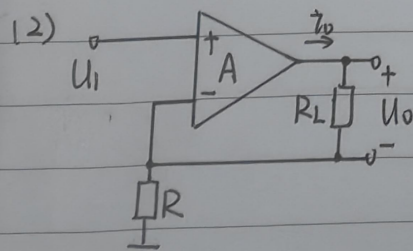
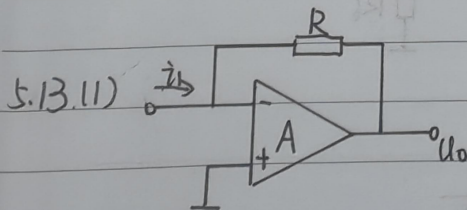
$$\therefore f_{H2} = f_{H3} = 10 f_{H1},$$

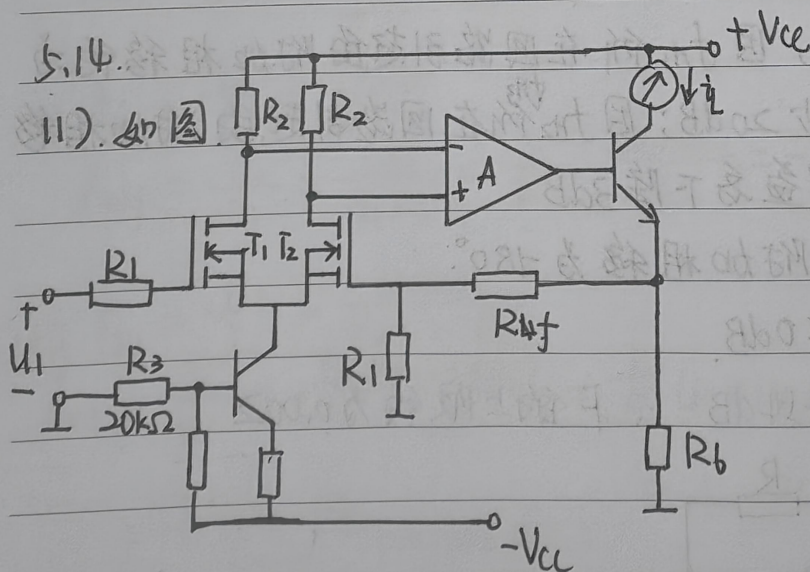
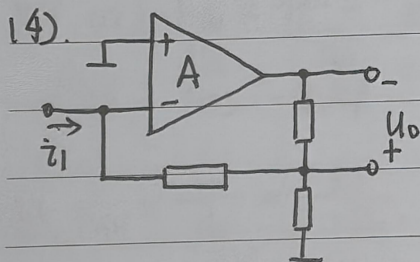
$\therefore f = 10^5 \text{ Hz}$ 时, 因 f_{H1} 所在回路引起的附加相移约为 -90° , 增益减少 20 dB ; 因 f_{H2} 所在回路引起的附加相移约为 -45° , 增益各下降 3 dB

$$\therefore |A| = 54 \text{ dB}, \text{ 附加相移为 } -180^\circ.$$

$$\text{为令 } 20 \lg |A\dot{F}| < 0 \text{ dB}$$

$$\text{则 } 20 \lg |\dot{F}| \leq -54 \text{ dB} \quad \therefore \dot{F} \text{ 的上限约为 } 0.002.$$





$$12) \dot{F} = \frac{U_t}{I_0} = \frac{R_1 R_6}{R_1 + R_2 + R_6}$$

$$\therefore \dot{A}_t \approx \frac{1}{\dot{F}} = \frac{R_1 + R_2 + R_6}{R_1 R_6}$$

$$\text{代入解得 } R_2 = 18.5 \text{ k}\Omega$$